MILITARY SPECIFICATION

COIL ASSEMBLY, TELEPHONE LOADING CU-260( )/G

1. SCOPE

1.1 Scope.- This specification covers Coil Assembly, Telephone Loading, CU-260( )/G.

2. APPLICABLE DOCUMENTS

2.1 Documents.- The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-571	Solder; Soft (Tin, Tin-Lead and Lead-Silver).
PPP-B-636	Boxes, Fiberboard.
PPP-B-640	Boxes, Fiberboard, Corrugated Triple Wall
PPP-P-291	Paperboard, Wrapping, Cushioning.
PPP-S-760	Strapping, Nonmetal
PPP-T-45	Tape, Gummed, Paper Reinforced.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper (For Carton Sealing).
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

FSC 5950

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MIL-P-116	Preservation, Methods of.
MIL-M-13231	Marking of Electronic Items.
MIL-F-14072	Finishes for Ground Signal Equipment.
MIL-B-43291	Boxes, Fiberboard, Corrugated, Double-Wall, Weather Resistant.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-170	Moisture Resistance Test Cycle for Ground Signal and Electronic Test Equipment.
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts.

DRAWINGS

ELECTRONICS COMMAND

SC-DL-74624	Telephone Loading Coil, CU-260( )/G.
SC-GL-57906	Gages for Telephone Loading Coil CU-260( )/G.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol shall be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Description.- Telephone Loading Coil Assembly CU-260( )/G is a metal cylinder approximately 5 inches long and 2 inches in diameter with a mating face at each end for connecting spiral-four type cable assemblies. The loading coil decreases cable attenuation in the 0-to-20 kilocycle frequency range. It is provided with end caps to protect the mating ends when not in use.

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3.2 Preproduction samples.- The contractor shall furnish preproduction samples as follows, provided the invitation for bids and contract requires samples.

12 ea Coil Assembly, Telephone Loading, CU-260( )/G.

3.3 Construction.- The loading coil shall be constructed in accordance with the requirements of this specification and Drawing and Data List SC-DL-74624.

3.4 Cast and molded parts.-

3.4.1 Casting.- Castings shall be of uniform quality and condition, and free from cracks, harmful shrinkage, porosity, gas holes, foreign matter, and other injurious defects. The surface of the castings shall be free from pits, parting lines, porous areas, fins, ridges, nodules, raised metal, and scale. All castings shall be completely cleaned prior to presentation for inspection. Castings shall not be plugged or welded, nor shall imperfections be filled in.

3.4.2 Molded parts.- Molded parts shall be uniform in quality, condition, and color. The molded parts shall be clean, smooth, free from porous areas, foreign materials, weak sections, bubbles, flash and any other injurious defects.

3.5 Cleaning.- After assembly, the equipment shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material.

3.6 Finish, protective.- Equipment shall be given protective finish in accordance with Specification MIL-F-14072 and the equipment drawings. (See 4.4).

3.7 Marking.- Marking shall conform to Specification MIL-M-13231. (See 4.4).

3.8 Electrical requirements.-

3.8.1 Dielectric strength.- The telephone loading coil assembly shall be capable of withstanding an ac potential of 1000 volts rms applied for one minute or 2500 volts rms applied for 30 seconds between each conductor in turn and the three remaining conductors connected to the case grounded. (See paragraph 4.7.1).

3.8.2 Insulation resistance.- Immediately after the dielectric strength test, the insulation resistance, measured between each conductor in turn and the remaining three conductors connected to the case grounded, shall be not less than 5000 megohms. (See paragraph 4.7.2).

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3.8.3 AC impedance.- Short circuit impedance shall be measured on each pair of the telephone loading coil assembly at a frequency of 20 kcps. The inductance obtained from this measurement shall be 6.0 millihenries, plus or minus four percent, and the ac resistance shall not exceed 3.77 ohms, at or corrected to 70° F. (See paragraph 4.7.3).

3.8.4 Continuity.- Electrical continuity shall exist between each male contact on either end and its corresponding female contact on the opposite end. (See. 4.7.4).

3.9 Mechanical requirements.-

3.9.1 Interchangeability.- The loading coils shall meet the gages specified in paragraph 4.8.1. Like units, assemblies, subassemblies and parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. Individual items shall not be hand-picked for fit. Reliance shall not be placed on any unspecified dimension, characteristic, etc.

3.9.2 Immersion.- There shall be no evidence of water into the assembly or between the assembly and the connectors after the 16 hour test specified in 4.8.2. The assemblies shall meet the requirements of Insulation resistance 3.8.2.

3.10 Moisture resistance.- When subjected to the test specified in 4.10, the loading coil shall meet the requirements of 3.8.1, 3.8.2 and 3.8.4 and there shall be no evidence of corrosion.

3.11 Workmanship.- The loading coil shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs herein. (See 4.9).

3.4 Cast and molded parts
3.5 Cleaning

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.3).

(b) Inspection covered by subsidiary documents. (See 4.4).

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5).

(2) Quality conformance inspection of preparation for delivery. (See 4.11).

4.3 Preproduction inspection.- This inspection shall consist of the preproduction inspection specified in Table I, the inspection specified in paragraph 4.4, and the inspection specified for group A and group B. (See Tables II and III, respectively).

Table I - Preproduction inspection

Inspection	Req. Para.	Insp. Para.
Moisture resistance	3.10	4.10

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4.4 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection before preparation for delivery:

Item	Where required
Finish	3.6
Marking	3.7

4.5 Quality conformance inspection of equipment before preparation for delivery.- The contractor, to demonstrate compliance with specified requirements, shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.2. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government—at its discretion—may perform all or any part of the specified inspection to verify the contractor's compliance with specified requirements. (See 6.4). Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection.- This inspection, including sampling, shall conform to Table II and the inspection procedures of Standard MIL-STD-105 using the general inspection levels. Group A inspection shall be performed in any order which is satisfactory to the Government.

Table II - Group A inspection

Inspection	Req. Para.	Insp. Para.	AQL Major	AQL Minor
Visual and mechanical	3.11	4.10	1.0%	4.0%
Electrical				
Dielectric strength	3.8.1	4.7.1	1.0% for the group	
Insulation resistance	3.8.2	4.7.2		
AC impedance	3.8.3	4.7.3		
Continuity	3.8.4	4.7.4		

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4.5.2 Group B inspection.- This inspection, including sampling, shall conform to Table III and to the special procedures for small-sample inspection of Standard MIL-STD-105. The AQL shall be 1.0 percent defective and the inspection level shall be S-4 for normal, tightened, and reduced inspection. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

Table III - Group B inspection

Inspection	Req. Para.	Insp. Para.
Interchangeability	3.9.1	4.8.1
Immersion	3.9.2	4.8.2

4.6 Preconditioning.- The contractor shall be permitted to precondition the assemblies prior to performing any electrical tests. The preconditioning shall consist of removing surface moisture from the assemblies by wiping, drying, or blowing. The application of heat shall be limited to three minutes or less. For those assemblies subjected to test specified in 4.8.2 Immersion and 4.10 Moisture resistance, without protective covers, a 24 hour waiting period in an ambient room temperature is permitted. However, if the contractor exercises this option of preconditioning, the contractor shall precondition all samples selected immediately prior to testing.

4.7 Electrical tests.-

4.7.1 Dielectric strength.- The equipment shall be tested in accordance with Method 301 of Standard MIL-STD-202 to determine compliance with the requirements of 3.8.1.

4.7.2 Insulation resistance.- The equipment shall be tested in accordance with Method 302, Condition B of Standard MIL-STD-202 to determine compliance with the requirements of 3.8.2.

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4.7.3 AC impedance.- Each pair of telephone loading coil assembly under test shall be tested for compliance with the requirements of paragraph 3.8.3 with a current of 0.5 to 5.0 milliamperes through the conductors.

4.7.4 Continuity.- Electrical continuity of the assembly shall be verified by a point-to-point test. (See 3.8.4).

4.8 Mechanical tests.-

4.8.1 Inspection for interchangeability.- The coil assemblies shall be gauged using the applicable gauges listed on Gage List SC-GL-57906 to determine conformance with the requirements of 3.9.1. When a mechanical value is not within specified or design limits, it shall be considered a major defect.

4.8.2 Immersion.- The Telephone Loading Coil CU-260( )/G shall be connected between contractor-furnished cable assemblies CX-1512( )/U and immersed in one foot of water at 18° to 32° C for 16 hours and meet the requirements of 3.9.2.

4.9 Visual and mechanical inspection.- The assemblies shall be examined for the defects listed in Table IV. (See 3.11).

Table IV - Classification of visual and mechanical defects

Classification	Defect
Major	Threads chipped, broken or stripped. Inability to rotate cup freely (cap). Rubber on contacts. Inability of coil to accept the protective cap. Substandard finish (Flaking, peeling, blisters, etc.) Contacts missing, broken or bent. Contact assembly parts not bonded together. Contact assembly cut or ripped. Any foreign objects or material preventing mating with connectors.
Minor	Missing parts. Finish not as specified. Marking incorrect. Abrasions on rubber parts.

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4.10 Moisture resistance.- The assemblies, unmated with and without protective covers in place and mated between cable assemblies CX-1512( )/U shall be placed in a humidity chamber and subjected to fifteen continuous 48 hour cycles in accordance with Standard MIL-STD-170. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL-STD-170. Insulation resistance measurements shall be made during the periods shown on the Standard, using Method 302, Condition A, of Standard MIL-STD-202. The assemblies shall meet the requirements of 3.10.

4.11 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C as specified. (See 6.2).

5.1.1 Level A.-

5.1.1.1 Cleaning.- Telephone Loading Coil CU-260( )/G shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying.- Telephone Loading Coil CU-260( )/G shall be dried in accordance with the applicable procedures of MIL-P-116.

5.1.1.3 Unit packaging.- Telephone Loading Coil CU-260( )/G shall be packaged Method III, as follows: Cushion each coil by wrapping in paperboard conforming to PPP-P-291 and secure with tape conforming to PPP-T-45, Type III.

5.1.1.3.1 Intermediate package.- A quantity of coils packaged as specified in 5.1.1.3 shall be placed within a close-fitting fiberboard box W6c conforming to PPP-B-636. Intermediate packages shall not exceed the gross weight of 20 pounds. Closure shall be made as specified in the appendix of the box specifications—in addition, seal all seams and joints with tape not less than two inches wide conforming to PPP-T-76.

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5.1.2 Level C.- Telephone Loading Coil CU-260( )/G shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing.- Packing shall be level A, B or C, as specified. (See 6.2)

5.2.1 Level A.- A quantity of Telephone Loading Coils CU-260( )/G, packaged as specified in 5.1, shall be packed within close-fitting fiberboard boxes conforming to PPP-B-636, Type CF, class weather-resistant, special requirements use, or PPP-B-640, class 2, grade A, style E or MIL-B-43291. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or non-metallic strapping conforming to PPP-T-97, Type IV and PPP-S-760, Type II, respectively: Selection of the material and application shall be in accordance with the appendix of the applicable box specification.

5.2.2 Level B.- Telephone Loading Coils CU-260( )/G, packaged as specified in 5.1, shall be packed as specified in 5.2.1 except that fiberboard boxes shall be class domestic conforming to PPP-B-636 and class 1, grade A, Style E conforming to PPP-B-640. Containers conforming to PPP-B-640 shall be reinforced as specified in 5.2.1 except non-metallic strapping and conforming to PPP-S-760 shall be Type 1.

5.2.3 Level C.- Telephone Loading Coil CU-260( )/G, packaged as specified in 5.1, shall be packed in shipping containers in a manner that will afford adequate protection against damage to the package and its contents during shipment from the supply source to the first receiving activity. Shipping containers shall comply with the rules and regulations of the common carrier as applicable to the mode of transportation.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use.- Coil Assembly, Telephone Loading CU-260( )/G is used with spiral-four cable assemblies at frequencies up to 20 kcps.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Type required.

(c) Level of packaging and level of packing required for shipment. (Level A, level B, or level C).

(d) Preproduction inspection.- Preproduction samples required as listed in paragraph 3.2.

(e) Place of final inspection.

6.3 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design: for example; CU-260W/G. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1).

6.4 Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

CUSTODIAN

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Preparing Activity

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Project No. 5950-A029

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SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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